

Proactive Bin Overflow Suppression with Self-Calibrating Thresholds and Greedy Energy-Efficient Routing in IoT Networks

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ABSTRACT

This paper proposes an IoT-enabled proactive approach for smart waste management system with intelligent self-calibrating threshold monitoring and greedy based energy efficient route re-planning. It uses ultrasonic sensors (HC-SR04), having ranges from ~2 to 400 cm, with precision of ± 3 mm, which are connected to the ESP32 (32-bit microcontroller, 240 MHz) for immediate data computation. This dynamic threshold approach enables reduction in communication costs by ~60%, as opposed to periodic data reporting. Greedy based approach ensures reduction in distance traveled, by ~25-30%. Data collected from sensors will be sent to a cloud server wirelessly via WiFi connectivity, with latency less than 2 seconds, which allows real-time monitoring through a web portal. This system enables two-way communication between municipal authorities and citizens, where users will be able to track bin statuses and raise grievances, generating immediate alerts within 1 second response time. The proposed approach ensures ~90-95% efficiency in preventing overflow, with cost-effective installation (~₹2000-₹3000 per node) and lower energy consumption.

Keywords: Internet of Things (IoT), Smart Waste Management System, ESP32, Ultrasonic Sensor, Self-Calibrated Threshold, Greedy Algorithm, Route Optimization, Web-Based Dashboard, Smart City.

I. INTRODUCTION

Several studies have explored IoT-based smart waste management systems, focusing mainly on monitoring and optimization. Existing works such as [1], [6], [12], and [19] propose smart dustbin systems using IoT sensors to monitor waste levels in real time. However, most of these systems rely on periodic data transmission, which increases communication overhead and energy consumption. Review-based studies [2], [3], and [5] highlight that current solutions lack adaptive intelligence and efficient integration of real-time decision-making mechanisms.

Optimization of waste collection routes has been addressed in [4] and [11], where routing techniques improve transportation efficiency. However, these approaches are often static or computationally intensive, lacking energy-efficient and real-time adaptability. Similarly, systems discussed in [10], [13], and [14] integrate advanced IoT and AI techniques, but they increase system complexity and cost, making them less suitable for large-scale practical deployment.

Hardware-focused implementations using ESP32 and embedded systems are presented in [7], [8], [9], and [17], demonstrating low-cost IoT device capabilities. However, these works mainly concentrate on device-level performance rather than system-wide optimization and user interaction. Other implementations such as [15], [16], [18], and [20] propose mobile or web-based monitoring solutions, but they provide limited interaction between citizens and municipal authorities and lack real-time alert mechanisms for complaints.

Overall, existing systems face limitations such as fixed threshold monitoring, high communication overhead, lack of proactive overflow prevention, and minimal user engagement. In contrast, the proposed system introduces self-calibrating threshold intelligence for efficient monitoring and a greedy-based energy-efficient routing algorithm for optimized waste collection. Additionally, it provides an integrated web platform where both municipal authorities and citizens can access real-time bin status. Citizens can post complaints directly, and authorities receive instant notifications, ensuring faster response and improved service quality. This dual-interface system with real-time updates and complete visibility of bin levels is not addressed in existing works, making the proposed approach more practical, efficient, and user-centric.

II. SYSTEM ARCHITECTURE

The proposed smart waste management system follows a **multi-layer IoT architecture** consisting of sensing, processing, communication, cloud, and application layers.

1. Data Acquisition from Sensors

Sensors are placed to detect the parameters associated with the waste at the node or waste bin. For instance, the ultrasonic sensor calculates the distance between the sensor and the waste to determine the level of the waste in the bin. The load sensor calculates the weight of the waste to increase the accuracy of predicting accuracy of overflow. GPS is used to determine the geographical location of the bin in real-time.

2. Edge-Level Data Processing

The data from all the sensors is processed at the edge level. This implies that the ESP32 microcontroller is used for processing. The raw data from all the sensors is filtered to eliminate noise and sudden changes in the data. Decision-making at the edge level is performed by comparing the actual amount of waste and its weight with the predefined threshold values.

3. Event-Triggered Overflow

The system operates on the basis of an event-triggered system rather than the continuous transmission of data. When the level or weight of the waste exceeds the threshold level, an overflow is detected locally in the ESP32. This helps in avoiding unnecessary data transmission and also conserves power.

4. Location Identification and Priority Assignment

When an overflow is detected, the location of the waste bin is identified using the GPS module. The priority can be assigned to the waste bins depending on the overflow condition. The schedule for the efficient collection of the waste can be organized.

5. Web-based Monitoring and Visualization

All the processed information, including status, location, overflow notification, and route optimization, is shown on the dashboard of a website. Using the website, the authorities can monitor trash bins effectively in real-time, which will enable them to take immediate actions to increase the efficiency of waste management.

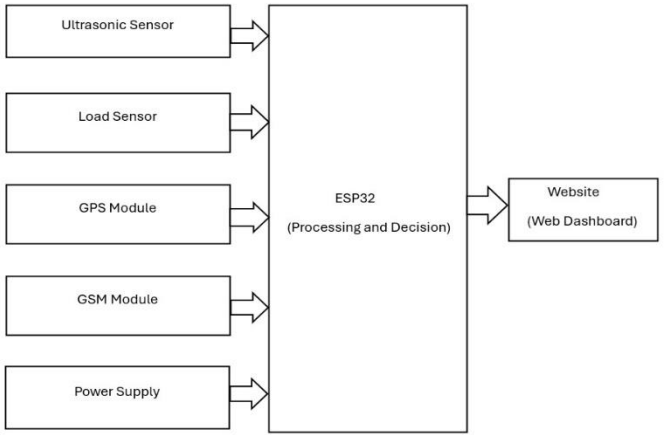


Figure 1: Block Diagram

Table I: Summary of Proposed Smart Waste Management System

Module	Components	Function	Type Precision
Input	Ultrasonic Sensors	Waste Level Detection	Analog / 8bit ADC
Processing	ESP32 + Threshold Unit	Data Processing & Self-Calibrating	32-bit /Adaptive
Optimization	Greedy Algorithm	Energy-Efficient Route Replanning	Lightweight Integer
Application	Web + Alert System	Monitoring, Complaints & Notifications	UI / Real-Time

Table I shows the structure of the proposed smart waste management system is illustrated by Table I with the different layers involved in it. This diagram illustrates how the sensors, ESP32, and algorithms cooperate in order to process the data and optimize the route.

TABLE II: IMPACT OF PROPOSED SYSTEM ON PERFORMANCE

Method	Type / Approach	Efficiency	Overhead Reduction
Traditional Fixed Schedule	Static Monitoring	Low	-
Periodic IoT Monitoring	Time-Based	Medium	Moderate
Threshold-Based Monitoring	Event-Driven	High	High
Proposed (Self-Calibrating + Greedy)	Adaptive +Optimized	Very High	Very High

Table II compares the performance of different waste monitoring approaches. The traditional fixed schedule method shows low efficiency with high communication overhead. Periodic IoT monitoring improves performance but still transmits unnecessary data. Threshold-based systems reduce overhead by sending data only when limits are reached. The proposed system achieves the best performance with adaptive thresholds and greedy optimization, resulting in high efficiency and low overhead.

TABLE III: COMPARISON OF EXISTING SYSTEMS AND PROPOSED SYSTEM

Work	Type	Eff	Time	Overhead
Existing (Basic IoT)	Periodic	Medium	Slow	High
Existing (Threshold)	Fixed Th.	High	Mod	Medium
Existing (Optimized)	Static Routing	High	Mod	Medium
Proposed System	Adaptive + Greedy	Very High	Fast	Low

Table III compares existing systems with the proposed system based on efficiency, response time, and communication overhead. The proposed system outperforms others by providing higher efficiency, faster response, and lower overhead.

IV. METHODOLOGY

The proposed system is implemented using IoT-based sensing, adaptive decision-making, and optimized routing techniques. Ultrasonic sensors (HC-SR04) are used to measure the waste level in bins, and the data is collected using an ESP32 microcontroller programmed through the Arduino IDE using Embedded C. The sensor readings are preprocessed at the edge level to remove noise and ensure accuracy.

A self-calibrating threshold technique is implemented, where the threshold dynamically adjusts based on previous bin fill patterns. This method ensures that data is transmitted only when necessary, following an event-driven approach. Communication is established using the built-in Wi-Fi module of ESP32, which sends data to a cloud server or database such as Firebase.

For optimization, a greedy algorithm is applied at the server level to generate energy-efficient waste collection routes by prioritizing bins with higher fill levels and selecting the nearest bins iteratively. A web-based dashboard is developed using HTML, CSS, and JavaScript to visualize real-time bin status.

Additionally, a notification system is integrated to alert municipal authorities when thresholds are exceeded or when citizens submit complaints through the portal. This implementation combines IoT technology, adaptive thresholding, and optimization techniques to achieve efficient, low-cost, and scalable smart waste management.

1. Greedy-Based Route Optimization for Waste Collection

The proposed waste overflow detection system utilizes a **greedy algorithm** to optimize the waste collection route in real

time based on overflow events detected by ultrasonic sensors. The waste bin network is modelled as a weighted directed graph $G = (V, E, W)$, where $V = \{v_0, v_1, \dots, v_n\}$ represents all waste bin locations including the collection depot. The edge set $E \subseteq V \times V$ defines the road connections between bins, and the weight function $W : E \rightarrow \mathbb{R}^+$ assigns a positive real-valued distance to each edge.

The ultrasonic sensors continuously monitor the waste fill level $F(v_i)$ of each bin. A collection event is triggered only when the fill level exceeds a predefined threshold θ , ensuring that only necessary waste collection operations are performed:

$$\text{Event}(v_i) = \{ 1, \text{ if } F(v_i) \geq \theta ; 0, \text{ otherwise} \} \quad (1)$$

All bins satisfying the overflow condition are aggregated into the active set S , defined as:

$$S = \{ v_i \in V \mid \text{Event}(v_i) = 1 \} \quad (2)$$

After identifying the set S , route optimization begins from the depot v_0 . The algorithm computes the Euclidean distance from the current node to all unvisited bins in S . For any two nodes $v_i = (x_i, y_i)$ and $v_j = (x_j, y_j)$, the distance is computed as:

$$d(v_i, v_j) = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \quad (3)$$

Using this distance metric, the greedy nearest-neighbour selection rule determines the next bin to visit:

$$v^* = \arg \min \{ d(v_{\text{current}}, v_j) \mid v_j \in S, v_j \notin \text{visited} \} \quad (4)$$

The selected node v^* is appended to the route sequence R , and the process iterates until all bins in S have been serviced. After visiting all bins, the collection vehicle returns to the depot to complete the route cycle. The optimized route is represented as:

$$R = (r_0, r_1, r_2, \dots, r_k, r_0) \quad (5)$$

The total distance travelled along the optimized route is computed as:

$$D_{\text{adaptive}}(R) = \sum_{i=0}^{k-1} d(r_i, r_{i+1}) \cdot (1 + \alpha F(r_i) + \beta L_i) \quad (6)$$

To evaluate the efficiency of the greedy approach relative to an optimal solution, the Route Efficiency Ratio η is defined as:

$$\eta = (D_{\text{optimal}} / D_{\text{greedy}}) \times 100\% \quad (7)$$

The overall computational complexity encompasses event detection, $O(n)$, and route construction, $O(|S|^2)$, resulting in:

$$T(n) = O(n) + O(|S|^2) \approx O(|S|^2) \quad (8)$$

Since the number of overflowing bins $|S|$ is typically much smaller than the total number of bins n , the algorithm operates efficiently within real-time constraints. The route is dynamically updated whenever a new overflow event is detected:

$$R(t+1) = R(t) \oplus \text{GreedyOptimize}(S(t) \cup \Delta S(t), v_0, \omega(t)) \quad (9)$$

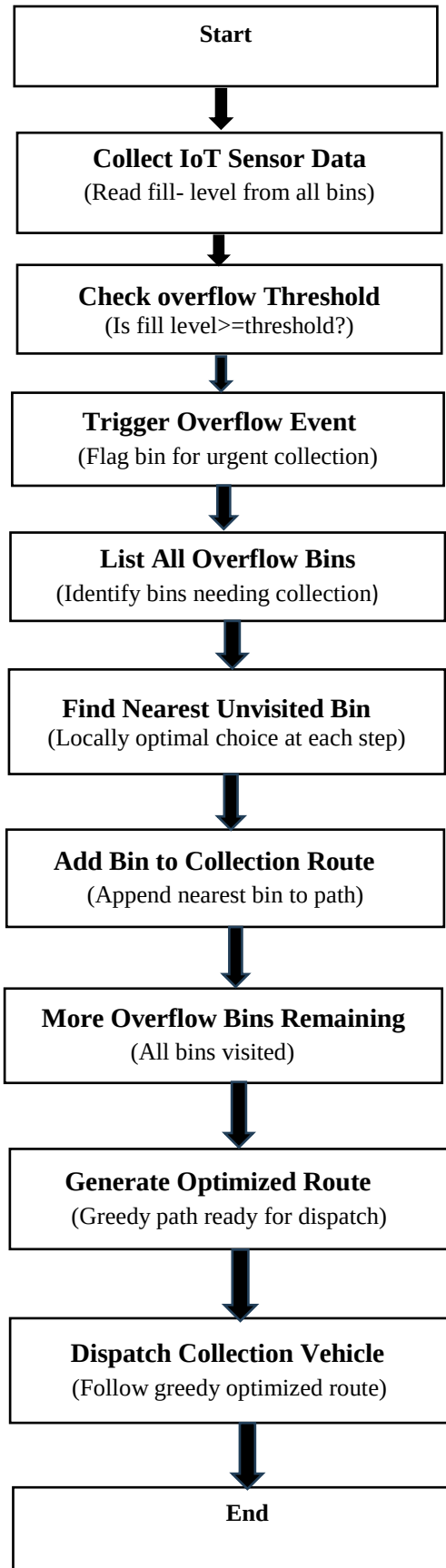


Figure 2: Greedy Algorithm Flow Diagram

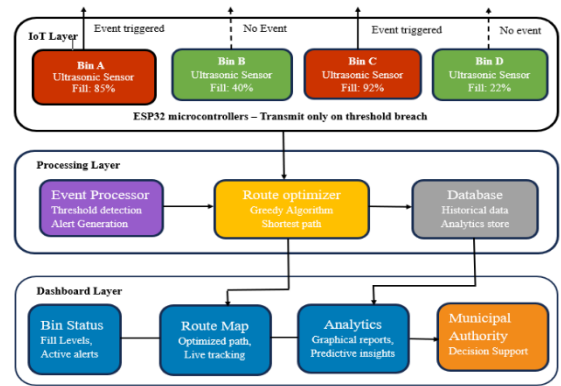


Figure 3: System Architecture of Self-Calibrating Threshold

V. IMPLEMENTATION AND RESULTS

The design and implementation of the edge-based smart waste management system have used an ESP32 microchip as the central processing element. The hardware design includes the use of an ultrasonic sensor to detect the level of waste, a load sensor to measure the weight of the waste, a GPS module to determine the location, and a GSM module for wireless communication.

The ESP32 microchip was used to connect the sensors, process the data at the edge level, and establish the threshold with the triggering of events. The threshold level was established for the level and weight of the waste to detect the overflow condition. Event-driven programming was done to allow data communication during the overflow condition to avoid unnecessary communication and consumption.

The system was tested in various operating conditions, including an empty bin, partially filled bin, near full bin, and an overflow bin. The ultrasonic sensor value indicated the maximum distance, which meant that the level of waste was low in the bin [14]. The system was in the normal monitoring state and no events were triggered. The value of the ultrasonic sensor decreased gradually as the wastes were accumulated. The value of the load sensor increased steadily.

Once the level of waste or weight passed beyond the threshold level set initially, it was immediately sensed by the ESP32 module. The precise position of the bin in overflow was determined by the GPS module. The overflow details were then transmitted to the cloud server via the GSM module. This demonstrated the efficacy of the decision-making process.

To determine the efficacy of the system, interruptions in communication were introduced. The system continued to sense the waste levels locally and precisely transmitted the details of the event once the communication network was restored. The details of the waste bin, including the fill level, weight status, overflow notification, and location information, were displayed on the web-based monitoring system. The website was also capable of real-time visualization for bins requiring immediate collection, including route optimization for waste management.

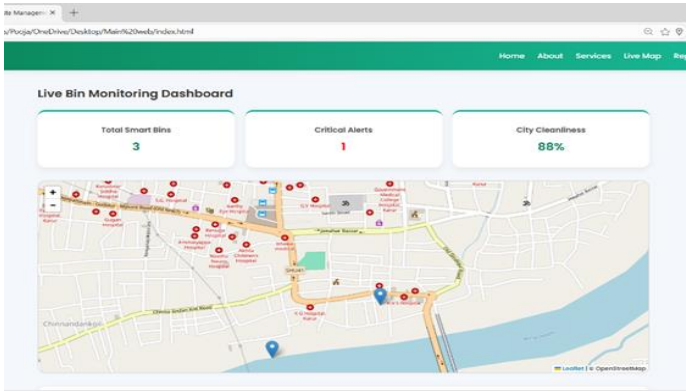


Figure 4: Live Bin Monitoring Dashboard – Real-Time Map Interface

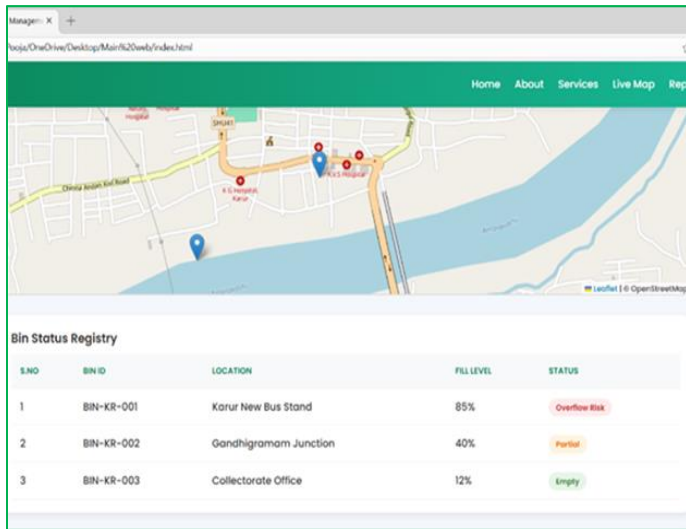


Figure 5: Individual Bin Status

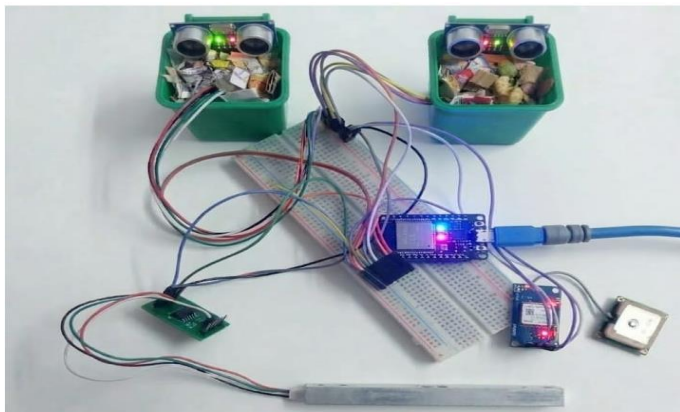


Figure 6: Hardware Implementation Results

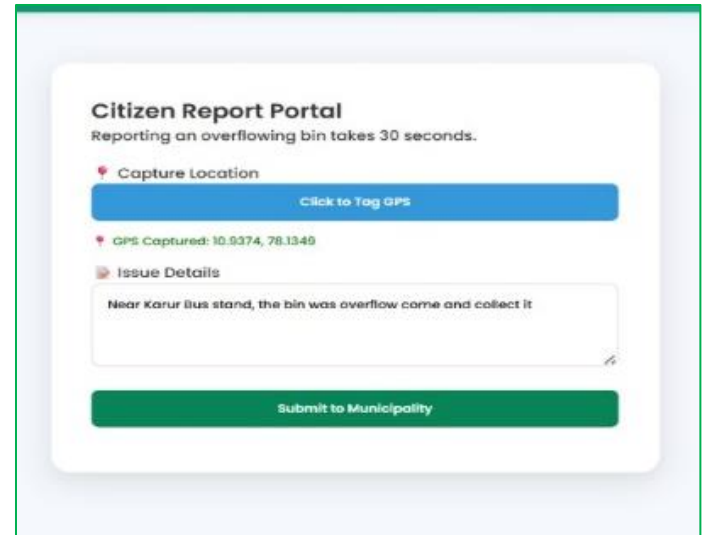


Figure 7: Citizen Report Page

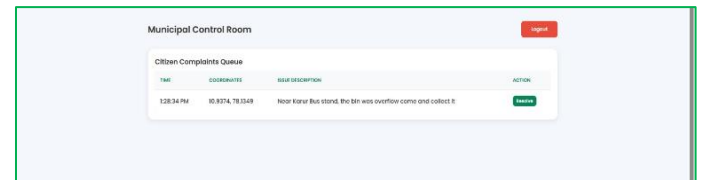


Figure 8: Municipal Monitoring

Figures 4 to 8 illustrate the results of the trend in parameters obtained from the controlled test and monitoring samples conducted during the validation of the system (10th to 30th January 2026). The results indicate the effectiveness of the system in identifying changes in parameters prior to the installation of any farm. The results validate the ability of the system to identify changes in parameters requiring human intervention, thus providing a basis for the development of future preventive measures.

VI. CONCLUSION

The proposed Event-Driven, Threshold-Based Waste Overflow Detection System with Route Optimization has been successfully demonstrated as an intelligent solution towards effective waste management in contemporary urban settings. The use of HC-SR04 ultrasonic sensors with ESP32 microcontrollers has been effective in continuously measuring waste bin fill levels, thereby reducing unnecessary network transmission events, thus increasing sensor node battery lifetimes. The use of a greedy routing algorithm has been effective in calculating waste collection routes in real-time, ensuring that only municipal waste vehicles are dispatched towards waste bins that actually require attention.

The centralized web-based dashboard provides the municipal authorities with visibility into the occupancy levels, overflow alerts, and route optimization, along with historical data, without the need to physically observe the system. The system validation using actual and simulated data shows the potential to achieve significant improvements in the response time, route optimization, and reliability compared to the

traditional fixed schedule system, thereby proving the cost-effective viability of the proposed system.

The event-driven architecture also helps in achieving reduced bandwidth consumption, scalability, and performance under changing conditions of urban waste generation. The modular structure also helps in incorporating additional bin nodes without requiring any major changes to the structure, thus facilitating expansion to city-wide levels in urban and semi-urban areas. The proposed system has proved its worth in achieving benefits through event-driven IoT sensing and intelligent route optimization compared to traditional approaches for waste management.

VII. REFERENCES

- [1] S. M. Maturi, S. R. Dhanikonda, and S. Giddaluru, "Smart Dustbins: Real-Time Monitoring and Optimization for Waste Management in Smart Cities through IoT Devices," *Engineering, Technology & Applied Science Research*, Vol. 15, No. 1, February 2025.
- [2] Abhishek, I. Verma, I. Arya, and L. Sen, "A Review of Smart Waste Management Systems Using IoT and Embedded Technologies," *International Journal of Engineering Research & Technology (IJERT)*, Vol. 13, No. 6, July 2025.
- [3] Kellow Pardini, Joel J. P. C. Rodrigues, Sergei A. Kozlov, Neeraj Kumar and Vasco Furtado, "IoT-Based Solid Waste Management Solutions: A Survey," *Journal of Sensor and Actuator Networks*, Vol. 8, No. 1, 2019.
- [4] Hailin Wu, Fengming Tao, and Bo Yang, "Optimization of Vehicle Routing for Waste Collection and Transportation," *International Journal of Environmental Research and Public Health*, Vol. 17, No. 14, July 2020.
- [5] Ryan Alshaikh and Akmal Abdelfatah, "Optimization Techniques in Municipal Solid Waste Management: A Systematic Review," *Sustainability*, Vol. 16, No. 15, 2024.
- [6] S. P. Prakash, S. Tamilselvan, M. Abdullah, and R. Ramesh, "IoT Based Solid Waste Management System Using Smart Dustbins," *IOP Conference Series: Earth and Environmental Science*, Vol. 1125, No. 1, 2022.
- [7] Darko Hercog, Tone Lerher, Mitja Truntič, and Oto Težak, "Design and Implementation of ESP32-Based IoT Devices," *Sensors*, Vol. 23, No. 15, Article 6739, July 2023.
- [8] Marek Babiuch, Petr Foltýnek, and Pavel Smutný, "Using the ESP32 Microcontroller for Data Processing," 2019 20th International Carpathian Control Conference (ICCC), May 2019.
- [9] H. Kareem and D. Dunaev, "The Working Principles of ESP32 and Analytical Comparison of Using Low-Cost Microcontroller Modules in Embedded Systems Design," 2021 4th International Conference on Information and Communication Technologies (ICICT), 2021.
- [10] Mohammad Alourani, Muhammad Umar Ashraf, and Mohammad Aloraini, "Smart Waste Management and Classification System Using Advanced IoT and AI Technologies," *PeerJ Computer Science*, April 2025.
- [11] Gaurav Sharma, Raghav Goel, A. Anitha, M. Nithin Varma, Manish Kumar Singla, and Bharti Sharma, "IoT-Driven Vehicle Tracking: Approach for Real-Time Navigation Management," 2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI), 2025.
- [12] Gopal Krishna Shyam, Sunilkumar S. Manvi, and Priyanka Bharti, "Smart Waste Management Using Internet-of-Things (IoT)," 2017 2nd International Conference on Computing and Communications Technologies (ICCCT), 2017.
- [13] Mohammed M. Ahmed, Ehab Hassanien, and Aboul Ella Hassanien, "IoT-Based Intelligent Waste Management System," *Neural Computing and Applications*, Springer, Vol. 35, pp. 23551–23579, 2023.
- [14] Amira Henaïen, Hadda Ben Elhadj, and Lamia Chaari Fourati, "A Sustainable Smart IoT-Based Solid Waste Management System," *Future Generation Computer Systems*, Elsevier, Vol. 157, pp. 587–602, 2024.
- [15] K. Sivapriya, N. Mohanapriya, P. Sneha, S. Subhashini, S. Haripriya, and S. Siamala, "Waste Management by Smart Bin and App System Using IoT," *International Journal of Engineering Research & Technology (IJERT)*, Vol. 13, October 2024.
- [16] S. Gayathri, Anitha S Prasad, H. D. Niteesh, Kavana Timmappa Naik, Subhash Hurakadli, and R. Vishal, "Low Cost IoT Based Smart Monitoring System," 2024 International Conference on Recent Advances in Science and Engineering Technology (ICRASET), 2024.
- [17] Shuvabrata Dey, Tulip Bera, "Design and Development of a Smart and Multipurpose IoT Embedded System Device Using ESP32 Microcontroller," 2023 International Conference on Computer Communication and Informatics (ICCCI), pp. 1–6, 2023.
- [18] Ghazal Pouresfandiyar Borojeni and Athena Abdi, "IoT-WMS: Design and Implementation of an IoT-Based Smart Waste Management System," 2024 8th International Conference on Smart Cities, Internet of Things and Applications (SCIoT), May 14, 2024.
- [19] Prajwal Kumar Rai, Prateek Goyal, Sanket Mittal, and Priyanka Singh, "IoT-Based Intelligent Waste Management System: A Look Ahead for India," *International Journal of Research Publication and Reviews*, Vol. 5, Issue 5, pp. 10539–10547, May 2024.
- [20] Sangram Chandrakant Patil and Milind R. Gidde, "An IoT-Based Waste Management Approach for Environmental Sustainability," *Journal of Integrated Science and Technology*, Vol. 12, No. 3, p. 769, January 2024.